5.1 What Are Ratios?

Lesson Introduction

<table>
<tr><td>Benchmark</td><td colspan="2">MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b, or $a:b$ where $b \neq 0$.</td><td>Learning Targets</td><td colspan="2">• I can write and interpret ratios.
• I can draw a diagram that represents a ratio and explain what the diagram means.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td colspan="2">MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.</td><td colspan="3">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="5">• What is a ratio?
• What are three ways to write ratios?
• Can a ratio be simplified?
• Explain if the ratios are the same if you switch the numbers.</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">In this first lesson on ratios, students are introduced to several different ways to describe a situation using ratio language. Initially, students sort figures by color, area, and a third attribute of their choice. Next, students learn how to write and interpret ratios and then practice these skills. In the 5.1.4 Guided Instruction, students learn to represent and interpret ratios with diagrams. Finally, students reflect on their ability to write, draw diagrams of, and interpret ratios.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>5.1.1 Warm-Up (~5 min.)</td><td>Sort figures by color, area, and a third attribute of their choice (SE p. 173)</td><td>5.1.4 Guided Instruction (10 min.)</td><td colspan="2">Represent ratios with diagrams (SE p. 176)</td></tr>
<tr><td>5.1.2 Guided Instruction (15 min.)</td><td>Write and interpret ratios (SE p. 174)</td><td>5.1.5 Collaboration (10-15 min.)</td><td colspan="2">Interpret ratios with diagrams (SE p. 177)</td></tr>
<tr><td>5.1.3 Try It (5-10 min.)</td><td>Students practice writing and interpreting ratios (SE p. 175)</td><td>5.1.6 Wrap-Up (~5 min.)</td><td colspan="2">Students reflect on their ability to write, draw diagrams of, and interpret ratios (SE p. 178)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
<tr><td colspan="2"><u>Key Terms</u>:
• Ratio

<u>Additional Terms:</u> N/A</td><td colspan="2">• (Optional) Note cards
• (Optional) Chart paper
• (Optional) Markers</td><td>• Consider creating an anchor chart.</td></tr>
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<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• Students may write ratios with no descriptive words or units. For example: $a\!:\!b$, and for some numbers, a and b.
• Students may not pay close attention to the order of the quantities, but the order of the quantities impacts the ratio. For example, $a\!:\!b$ is not the same as $b\!:\!a$, for some numbers a and b.
• Students may think ratios and fractions are the same when they are not. Fractions compare part-to-whole relationships while ratios can compare part-to-part or part-to-whole relationships.</td><td>• In the Warm-Up, the teacher should differentiate between the small L-shaped figures and the large L-shaped figures.
• Writing a ratio as $a\!:\!b$, for some numbers a and b is a good start, but part of writing a ratio is stating what those numbers mean. Draw students' attention to the sentence stems in the task statement and encourage them to use those words.
• Create a reference poster and allow students to create reference cards with instructions and examples to aid them in writing and interpreting ratios represented with diagrams.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
The Think-Pair-Share routine can be used in components 5.1.1-5.1.4 of the lesson. Initially, have students attempt the questions individually. Next, pair students up and ask them to discuss their strategies, reasoning, and answers. Tell students to resolve any differences in their answers. This routine helps students build their confidence and understanding.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Throughout the lesson, students connect mathematical concepts to everyday experience. They model real-world scenarios with ratios. They also analyze and describe diagrams that represent real-world ratio relationships. Students must critically think about the order of the quantities and whether they are asked to find a part-to-part or a part-to-whole ratio.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider having students create a reference card with an explanation of how to write and interpret ratios represented in diagrams. Encourage students to include at least one example of each process. Allow students to use their reference cards during the lesson. The diagrams of the ratios provide a visual entry point. Structuring components with the Think-Pair-Share routine provides an auditory entry point.
Mark-Up Text: Notice the use of Mark-Up on a number line in the Wrap-Up. Consider utilizing Mark-Up during lesson components for students to notate if they are uncertain about a portion of the lesson or if they feel confident answering the questions.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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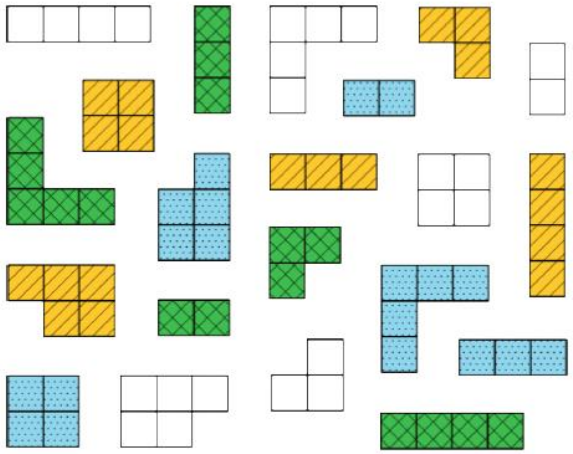
5.1.1 Warm-Up: What Kind and How Many?

Component Overview: *In this Warm-Up, students compare figures and sort them into categories. The first two questions are straightforward to allow all students access to the component and prime them to think about other ways the figures can be sorted. Students create their own categories in the third question and explain their reasoning.*

5.1.1 Warm-Up: What Kind and How Many?

A set of figures with different colors, shapes, and sizes is shown.



The figures can be sorted using different categories.

1. Complete the table by determining the number of groups for each method of sorting. For the last row, determine a third way the figures could be sorted and how many groups the sorting method would create.

Sorting Method	Number of Groups Created
Sorted by color	4
Sorted by area	4
Sorted by *Answers will vary* *Sorted by shape*	*Answers will vary* 7

Think-Pair-Share
This Warm-Up provides an opportunity to engage students in discourse to discover the multiple ways the figures can be sorted. Engage students in a Think-Pair-Share after completing the table. Have students explain how they sorted the figures. Emphasize that the important thing is to describe the way they sorted them clearly enough, so everyone agrees it is a reasonable way to sort them. Tell students we will be looking at different ways of seeing the same set of objects in the next component.

5.1.2 Guided Instruction: Writing and Interpreting Ratios

Component Overview: *In this exploration, students first complete statements to describe the relationships between the figures in the Warm-Up. Students learn that they can express the ratio of a to b three ways:* $\boldsymbol{a}:\boldsymbol{b}$, $\boldsymbol{a}$ *to* $\boldsymbol{b}$, *or* $\frac{a}{b}$. *Finally, students write and interpret ratios measuring the amount of chai powder used in recipes for chai tea.*

5.1.2 Guided Instruction: Writing and Interpreting Ratios

The figures from the Warm-Up can be described using ratio language.

A ***ratio*** is a comparison of two or more quantities.

1. Complete the statements to describe the relationships between the figures from the Warm-Up.

   a. There are 5 yellow figures for every 5 green figures.

   b. For every 3 figures with an area of 2 square units, there are 6 figures with an area of 5 square units.

   c. The ratio of square-shaped figures to L-shaped figures is 3 to 6.

Ratios can be written in three ways. The ratio of blue figures to white figures is expressed in all three ways below.

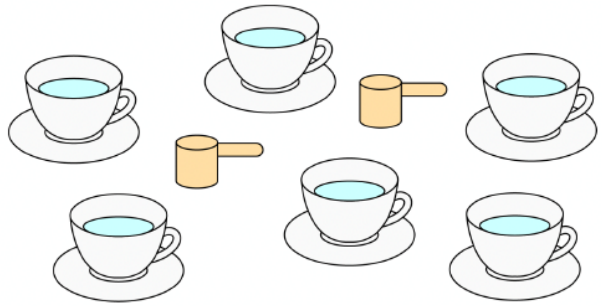
Method for Writing Ratios	The Ratio of Blue Figures to White Figures
Using a colon (:)	$5:6$
Using the word "to"	5 to 6
Using a fraction bar	$\frac{5}{6}$

This Guided Instruction is intentionally scaffolded, as this is the first-time students are working with ratios. Model your approach and language in questions 1 and 2 before giving students an opportunity to try it on their own in question 3.

Consider engaging students with a hook here to give them a creative opportunity to make sense of ratios — What are two quantities you would love to have in a ratio of $5:2$ *but hate to have in a ratio of* $2:5$*?*

5.1.2 Guided Instruction: Writing and Interpreting Ratios (continued)

2. Avril makes a tea called chai using a pre-made powdered mix. The box instructions say to mix 2 scoops of chai powder with 6 cups of hot water.



Complete each description of the collection by writing a number in each box.

a. The ratio of chai mix scoops to cups of water is

$\boxed{2} : \boxed{6}$.

b. The ratio of cups of water to chai mix scoops is

$\dfrac{\boxed{6}}{\boxed{2}}$.

Students may confuse writing the ratio of scoops to cups of water and cups of water to scoops. You may see students writing both ratios as $2:6$ or $6:2$, or students may mix up the numbers for each answer.
How many scoops of chai are needed to make the tea? What box should this go in if we are describing the ratio of scoops to cups of water?
How many cups of water are needed? What box should this go in if we are describing the ratio of cups of water to scoops?
Will the ratio of scoops to cups of water be the same as the ratio of cups of water to scoops? Why not?

5.1.2 Guided Instruction: Writing and Interpreting Ratios (continued)

3. The relationship between chai mix and water can also be described using other numbers. Consider the rearranged image.

Complete the statements.

a. There are 3 cups of water per scoop of chai mix.

b. There is $\frac{1}{3}$ scoop of chai mix for every cup of water.

Notice the fraction bar for question 3b. Notice how students articulate their responses, as this is not actually a fraction of a part:whole relationship but instead a ratio representing a part:part relationship.
Consider pausing to ask students whether something represents a part:part or part:whole relationship.

The order of the ratio is important, as it relates specifically to the quantities being described. The order used in the written description must match the order of the values that are being compared.

The built-in partner and small-group discussions provide an entry point for auditory learners. To engage visual learners, encourage students to label the ratio boxes with different colors, one for scoops and one for cups of water. This can help students keep track of how to write the ratio correctly.

4. Discuss with your partner why the ratio of chai mix scoops to cups of water is not equivalent to the ratio of cups of water to chai mix scoops.
Answers will vary. There are a different number of chai mix scoops and cups of water. The order matters when describing ratios. When referring to values like 1 and 3, I have to keep them in order so I know which value goes with which unit.

Consider providing sentence stems for additional scaffolding for question 4.
The ratios are not equivalent because there are a different number of ______ and ______.
The order (does/does not) matter when writing ratios.
When I write ratios, I need to pay attention to _____.

5.1.3 Try It: Writing and Interpreting Ratios

Component Overview: *Students work with a partner to write and interpret ratios in a given context.*

5.1.3 Try It: Writing and Interpreting Ratios

In a fruit bowl, there are 6 oranges, 4 apples, and 2 plums.

1. Complete the statements to describe the contents of the fruit bowl.

   a. The ratio of oranges to apples is 6 : 4.

   b. The ratio of plums to apples is 2 to 4.

   c. For every 4 apples, there are 2 plums.

   d. For every 3 oranges, there is one
   - ○ apple.
   - ○ plum.

2. Explain how you determined the answer for Part d using words or a sketch.

   Divide the 6 oranges into equal groups of 3. There are 2 groups of 3 oranges. Then, divide the 4 apples into 2 groups, which resulted in 2 apples per group. Next, divide the 2 plums into 2 groups, which resulted in 1 plum per group.

6 Oranges		
4 Apples		
2 Plums		

During this component, students should be engaging in informal discussions to find the ratios of each type of fruit in the bowl. Circulate the room to listen for how students are determining each ratio and continue to use prompting questions to clarify misconceptions as they arise.

- *Where should you write the number of oranges if it is asking for the ratio of oranges to apples?*
- *How can you determine the ratio of 3 oranges to apples or plums? Is there a way to represent this visually?*
- *How can you show these ratios using a colon or fraction bar?*

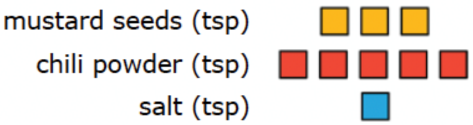
5.1.4 Guided Instruction: Representing Ratios With Diagrams

Component Overview: *Guide students through this component as they consider how to use ratios to represent part:part and part:whole relationships. Students work with a partner to represent the relationships between different ingredients in a recipe using ratios to represent the part to part and part to whole relationships.*

5.1.4 Guided Instruction: Representing Ratios with Diagrams

When solving problems involving ratios, it can be helpful to use diagrams and number lines to represent the relationships described.

1. A recipe for 1 batch of spice mix says, "Combine 3 teaspoons (tsp) of mustard seeds, 5 teaspoons of chili powder, and 1 teaspoon of salt." Below is a diagram to represent this situation.



How many teaspoons of spices are in 1 batch of the mix?
9 teaspoons

Ratios can be used to describe part-to-part relationships, where one part is being compared to another, and part-to-whole relationships, where one part is being compared to the entire group.

2. Complete the statements in the table.

Type of Relationship	Example Statements
part-to-part	The ratio of tsp of chili powder to tsp of mustard seeds is 5:3. There are 3 teaspoons of mustard seeds per tsp of salt.
part-to-whole	The ratio of teaspoons of chili powder to teaspoons of spice mix is 5 to 9. For every 1 teaspoon(s) of salt, 9 teaspoon(s) of spice mix can be made.

3. Discuss with your partner what is meant by "whole" in part-to-whole.

Consider having students create a reference card with an explanation of how to write and interpret ratios represented in diagrams. Encourage students to include at least one example of each process. Allow students to use their reference cards as a personal anchor chart during the lesson.

Compare and Connect
Consider using the Compare and Connect routine when students discuss their answers with their partners.

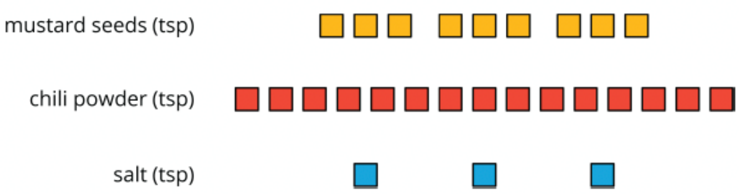
- *What is the same?*
- *What is different?*
- *What is meant by "whole" (question 3) and what does that make you think about writing ratios?*

5.1.5 Collaboration: Interpreting Ratios With Diagrams

Component Overview: *Students collaboratively practice writing and interpreting ratios of a spice mix recipe and a paint mixture that are represented with diagrams. The diagrams of the ratios provide a visual entry point. Structuring components with the Think-Pair-Share routine provides an auditory entry point.*

5.1.5 Collaboration: Interpreting Ratios with Diagrams

1. The diagram shows the ingredients Luca mixed to make the same spice mix recipe.



Work with your partner to determine how many batches of spice mix are represented by the diagram. Explain or show your reasoning.

Luca made 3 batches of spice mix. In Luca's mix, there are 9 teaspoons of mustard seeds. Each batch is to contain only 3 teaspoons of mustard seeds. 9 teaspoons of mustard seeds in Luca's mix $\div 3$ teaspoons in the recipe $= 3$ batches of spice mix.

	Mustard seeds	**Chili Powder**	**Salt**
tsp in Luca's mix	9	15	3
Recipe mix	3	5	1
Luca's mix ÷ Recipe mix	$9 \div 3$	$15 \div 5$	$3 \div 1$
Number of Batches of Spice Mix	$= 3$	$= 3$	$= 3$

The table is scaffolded in a way that guides students toward understanding the number of batches of spice mix represented by the diagram. To further guide students toward understanding, consider asking the following questions: How can you determine the number of batches? How does the table help us find the number of batches of spice mix? Is there a way to write a ratio to show this?

5.1.5 Collaboration: Interpreting Ratios With Diagrams (continued)

2. When mixing paint colors, the same shade is produced when the same ratios of each color are used to mix the paint. The diagram represents 4 batches of light green paint that Mabelise mixed.

white paint (cups) □□□□□□□□□□□□

green paint (cups) ■■■■■■■■■■■■■■■■

Use the diagram to answer the following questions with your partner.

a. Draw a diagram that represents 1 batch of the same shade of light green paint.
Divide the number of cups of each paint color by 4.

white paint (cups)	□□□	□□□	□□□	□□□
green paint (cups)	■■■■	■■■■	■■■■	■■■■

b. Complete the statements to describe the relationships.

- For every 3 cups of white paint, there are 4 cups of green paint in the mix.
- The ratio of ○ white paint ○ green paint to ○ white paint ○ green paint is $\frac{4}{3}$.
- The ratio of white paint to the light green mixture is 3 to 7.
- There are 8 cups of green paint used for every 14 cups of light green paint that is mixed.

This component provides visual and auditory entry points for students through the use of diagrams and partner discussion. Encourage students to use the visual scaffolds to determine the different ratios in the table and connect them to the abstract representations of ratios using colons and fraction bars.

Partner pairs should make connections between the visual representation of the paint and writing this as a ratio. Students should be able to articulate the ratio of green to white paint through connection to the visual but may struggle to determine the ratio of white to light green. Encourage students to circle each batch of light green in the table to provide a visual representation of light green paint.

5.1.6 Wrap-Up: Reflection

Component Overview: *Students reflect on their understanding and confidence in writing and interpreting ratios, as well as drawing and explaining diagrams that represent ratios.*

5.1.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.

1. I can write and interpret ratios.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

Answers will vary. Students may mark that they are "getting there but need more practice."

2. I can draw a diagram that represents a ratio and explain what a diagram means.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

Answers will vary. Students may mark that they are "getting there but need more practice."

Consider having students provide justification or evidence, either through written expression or visual demonstration.